## 4.2 Functions of propeller shaft

- It transmits the engine's power from gear box to the differential unit smoothly.
- It transmits the rotational power from the gear box at different angles as per requirement.
- According to the load of the vehicle, the length of the propeller shaft varies and transmits the power through slip joints.
- When the vehicle is running, it completely withstands the vibration occurring in the transmission system.

## 4.3 Types of propeller shaft drive

The operation of propeller shaft is classified in to two types. They are

1. Hotchkiss Drive
2. Torque tube Drive

### 4.3.1 Hotchkiss drive

#### Construction

In this drive, the leaf springs are tightly fixed at the center of the axis of rear axle. The back end of the spring is attached with Shackle (movable). The front end is attached to the chassis frame with anchor pin (fixed). The two universal joints and a sliding joint were connected to the Propeller Shaft. The first universal joint is connected to the gear box shaft and second universal joint is attached with the bevel pinion shaft of the differential unit. The construction of the Hotchkiss Drive is shown in Figure 4.3.1.

#### Working Principle

When the vehicle goes forward with more loads, the front end of the propeller shaft elongates and the rear end of the leaf spring is pressed and the front part of the spring is elongated. This happens due to the front and backward movements of the universal joint and sliding joint located at the front end of the propeller shaft. Due to this the vehicle moves easily on the road.

This Hotchkiss drive is used in most of the vehicles. This type of drive makes the vehicle to move smoothly even on the roads with ups and down.

#### Advantages

1. Weight is less
2. Simple in construction
3. It does not make more noise.

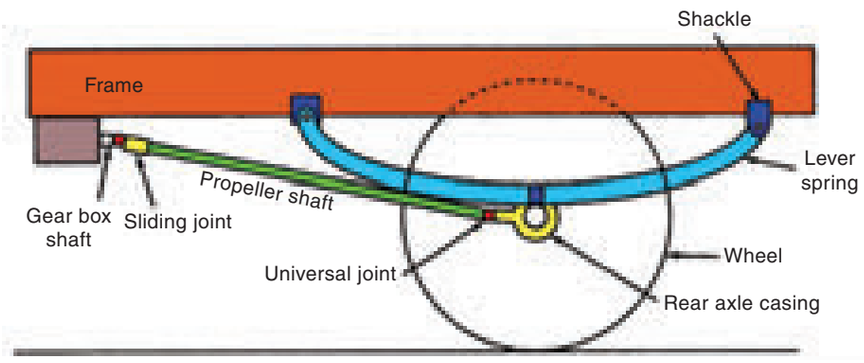



**Figure 4.3.1** Hotchkiss drive



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